

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 27)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

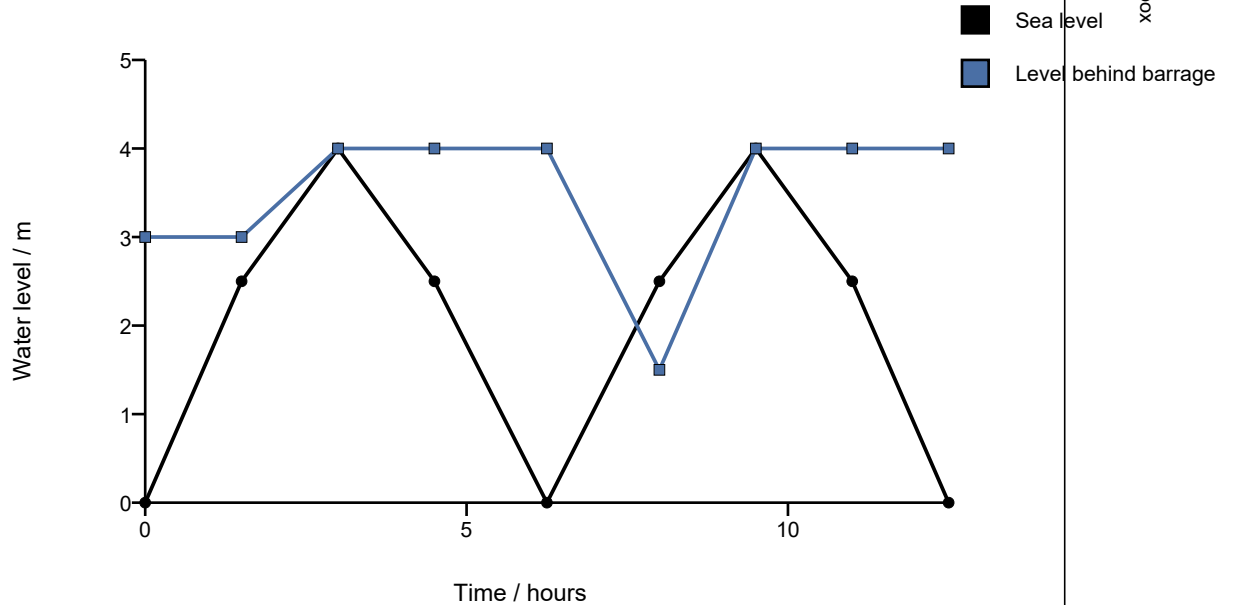
Question	Mark
1	
2	
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7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows the water level inside and outside a tidal barrage over one 12.5-hour tidal cycle and the periods when electricity is generated.

Figure 1



0 1 . 1

Use Figure 1 to explain when electricity is generated by the barrage.

[2 marks]

0 1 . 2

Explain why tidal power is intermittent but predictable.

[2 marks]

0 1 . 3

Explain the difference between spring and neap tides and why this affects output.

[2 marks]

0 1 . 4

Explain how a tidal lagoon can be used to generate electricity when there is no tidal flow.

[2 marks]

0 1 . 5

Give **two** coastal features that make a site suitable for tidal power.

[2 marks]

1

2

10

0 2 . 1

Describe **three** effects of noise on human health other than hearing damage.

[3 marks]

1

2

3

0 2 . 2

Explain how noise can damage non-living objects.

[3 marks]

0 2 . 3

Explain how noise can reduce the breeding success of birds.

[2 marks]

0 2 . 4

Explain why military sonar is a threat to cetaceans.

[2 marks]

10

0 3 . 1

Explain how each of the following human activities alters processes in the hydrological cycle.

[6 marks]

Deforestation

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Urban development

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Agriculture

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03 . 2

Describe **two** ways in which global climate change alters the hydrological cycle.

[2 marks]

1

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2

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03 . 3

Explain why the impact of industrial and domestic water use on the hydrological cycle is usually local.

[2 marks]

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10

0 4 . 1

Explain why oil and gas trapped in shale cannot be extracted by conventional drilling and describe how hydraulic fracturing (fracking) allows extraction.

[3 marks]

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0 4 . 2

Describe **four** environmental concerns about fracking and suggest how they may be reduced.

[5 marks]

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0 4 . 3

Explain why tar sands and oil shales are not included in most estimates of oil reserves despite containing more oil than conventional crude oil.

[2 marks]

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10

05

Mercury released into a lake from a factory was in the inorganic form. Fish in the lake were later found to contain high concentrations of methyl mercury, an organic form.

05

. 1

Suggest how the inorganic mercury was converted to methyl mercury.

[2 marks]

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05

. 2

Explain why methyl mercury is a more hazardous pollutant than inorganic mercury.

[3 marks]

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05

. 3

Explain why the health risk from eating fish from the lake may remain for many years after the discharges stop.

[2 marks]

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05

. 4

Suggest **three** factors that should be included in a Critical Pathway Analysis of a proposed mercury discharge.

[3 marks]

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1

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10

0 6 . 1 Give **three** sources of oil pollution of the sea other than tanker accidents.

[3 marks]

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2

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0 6 . 2 Explain how an oil spill at sea can kill seabirds.

[3 marks]

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0 6 . 3 Explain **two** ways in which an oil slick can harm organisms that live in the water beneath it.

[2 marks]

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06 . 4

Explain why oil pollution is most serious when it occurs inshore or in an enclosed body of water.

[2 marks]

10

07 . 1

Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence

☐

B Low liposolubility and low persistence

☐

C High liposolubility and low persistence

☐

D Low liposolubility and high persistence

☐

07 . 2

Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

07 . 3

Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

5

08 . 1

Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

- A Desalinated seawater
 B River water
 C Rainwater catchment
 D Newly discovered aquifers

☐
☐
☐
☐

08 . 2

Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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08 . 3

Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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5

09

Table 1 shows the energy used per passenger-kilometre by different forms of transport when carrying a typical load.

Table 1

Mode of transport	Typical occupancy	Energy use / MJ per passenger-km
Car (petrol, 1 occupant)	1	2.9
Car (petrol, 4 occupants)	4	0.7
Bus (diesel)	25	0.6
Electric train	200	0.3
Bicycle	1	0.06
Short-haul aircraft	120	2.4

09 . 1

Calculate the percentage reduction in energy per passenger-km achieved by carrying four people in a car instead of one.

[1 mark]

Answer %

09 . 2

Explain why bulk transport (buses, trains) uses less energy per passenger-km than cars.

[2 marks]

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09 . 3

Suggest **three** reasons why people continue to use cars despite the energy advantages of public transport.

[3 marks]

1

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2

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3

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09 . 4

Explain how integrated transport systems and Active Traffic Management reduce energy use.

[4 marks]

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10

1 0

A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

1 0

. 1

Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

1 0

. 2

The test gave $p = 0.01$. State the conclusion the student should draw.

[2 marks]

1 0

. 3

Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

Do not write outside the box

Extra space

15

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

[illegible]

END OF QUESTIONS